Do Now · retrieval

C# Login Project · Complete and Test

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In a seven-bucket hash table, a collision occurs at bucket 6. Which bucket is checked next by linear probing, and why?

Q2

Predict the output of Console.WriteLine((age >= 16) && hasPermission); when age = 17 and hasPermission = false. Explain.

Q3

In a binary search tree, a new key is greater than the root but smaller than the root’s right child. Where will it go if that right child has no left child?

Q4

A print server must process jobs in the order submitted. Choose a suitable data structure and explain why a stack would be unsuitable.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Your goal: a reliable demonstration system — Students will join registration and lookup so the program recognises existing usernames, refuses duplicates and performs login checks.

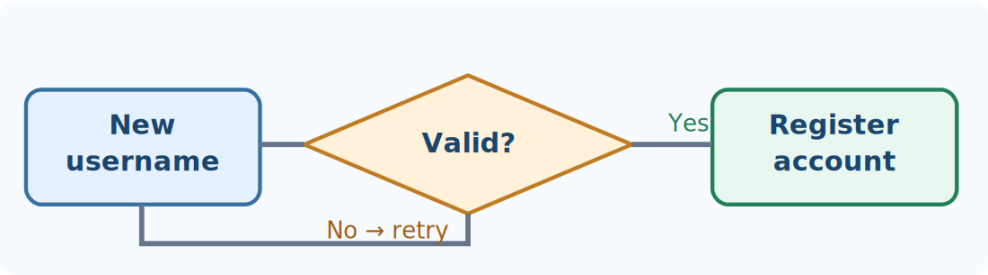
Core idea 2

Validation before storage — Reject blank usernames, duplicates and registrations when all buckets are full. Show a meaningful reason rather than crashing.

Core idea 3

Separate locating an account from checking a password — First use the hash table to find the username. Then compare an invented classroom demo password. These are separate operations.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Security boundary — The simple hash function here is unsuitable for real authentication. Production systems need established salted password-hashing libraries and secure storage.

Core idea 5

Synoptic retrieval — Exchange MCQs, answer independently, discuss distractors, then correct any technical errors. Cover the full OCR data-structures checklist.

Vocabulary · use the correct technical terms

Confirm data meets input rules. — Confirm data meets input rules.

A repeated key that must not be inserted twice. — A repeated key that must not be inserted twice.

Check a value at a limit, such as a full table. — Check a value at a limit, such as a full table.

Use invalid inputs to check handling. — Use invalid inputs to check handling.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Model validation before registration

Follow the worked method

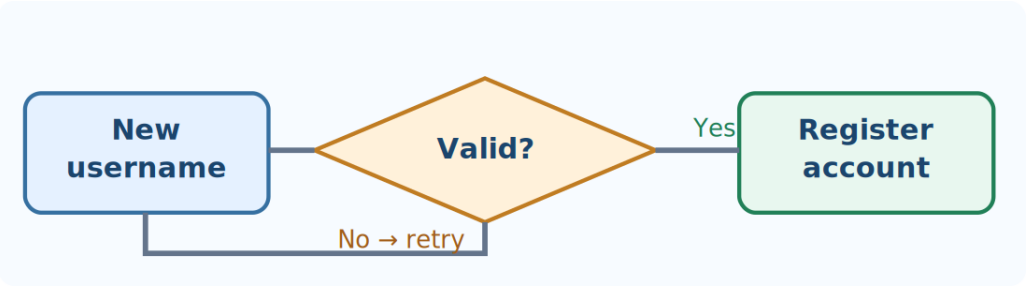
1. Check blank input — CODE:

```
Console.Write("New username: ");
string username = (Console.ReadLine() ?? "").Trim();
if (username.Length == 0)
{
Console.WriteLine("Username required");
}
```

2. Check whether the username exists — Use the lookup function from Lesson 033. Do not register duplicates, including usernames moved by probing.

3. Enforce a size limit — Track how many table buckets are occupied. If all seven are full, refuse a new account gracefully.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain your solution first
Which checks must happen before insertion?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

CODING 1 · COMPLETE REGISTER

Extend the Lesson 033 table with registration, blank-name checks, duplicate-name rejection and full-table handling. Keep existing hash and lookup functions.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

SYNOPTIC MCQ ASSESSMENT

Exchange student-written MCQs and answer independently. Cover arrays up to 3D, records, lists, tuples, linked lists, directed/undirected graphs, stacks, queues, trees, BSTs and hash tables. Identify and correct ambiguous distractors.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Two usernames producing the same hash index must always overwrite one another.

Use key vocabulary from memory

1. Explain Confirm data meets input rules. in your own words.

2. Explain A repeated key that must not be inserted twice. in your own words.

3. Explain Check a value at a limit, such as a full table. in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.